

Intel 80486 Microprocessor

1 Introduction

The **Intel 80486** is a 32-bit microprocessor and a major enhancement over the 80386 processor. It integrates the floating-point unit, cache memory, and pipeline architecture on a single chip, resulting in significantly improved performance.

2 Basic Architecture

- 32-bit processor
- Divided into three major sections:
 - Bus Interface Unit (BIU)
 - Execution and Control Unit (EU)
 - Floating Point Unit (FPU)

3 Bus Interface Unit (BIU)

- Organizes all bus activities
- Address driver connected to:
 - Internal 32-bit address output of cache
 - 32-bit system bus
- Data bus transceivers connect internal 32-bit data bus to system bus
- Write data buffer consists of a queue of four 80-bit registers

3.1 Parity Generation and Control Unit

- Generates parity bits
- Performs parity checking during processor operation
- Data bus width: 32 bits
- Address bus width: 32 bits
- Control bus width: 16 bits

4 Execution and Control Unit

4.1 Boundary Scan Control Unit

- Performs boundary scan test operations
- Ensures correct operation of all motherboard components

4.2 Prefetcher Unit

- Fetches instructions from memory
- Stores instructions in a 32-byte code queue

4.3 Instruction Decoder

- Receives instruction codes from the code queue
- Decodes instructions sequentially
- Sends decoded output to the control unit

4.4 Protection Unit

- Verifies protection rules before instruction execution
- Generates appropriate exceptions on violation

4.5 Common Exceptions

- Page Fault
- General Protection Fault
- Segment Not Present
- Stack Fault
- Invalid Opcode

5 Control ROM and Barrel Shifter

5.1 Control ROM

- Stores microprogram (microcode)
- Microcode controls:
 - Internal CPU operations
 - Data movement between registers
 - Activation of ALU, shifter, FPU, and memory interface
 - Execution of complex instructions

5.2 Barrel Shifter

- Performs shift and rotate operations
- Uses multiple parallel paths
- Allows shifts of up to 31 bits in a single operation

6 Floating Point Unit (FPU)

- First processor with in-built 80387 floating-point unit
- Used for mathematical data processing
- Operates much faster than the ALU
- Communicates with BIU under MMU control
- Uses a 64-bit internal data bus

7 Register Bank and EFLAGS

- Registers perform standard CPU operations
- FS and GS segment registers point to:
 - Process control block
 - Exception handlers
 - System tables
 - Kernel memory

7.1 Important EFLAGS Bits

- AC – Alignment Check
- IOPL – I/O Privilege Level
- NT – Nested Task
- RF – Resume Flag
- VM – Virtual 8086 Mode

8 Key Features of Intel 80486

- Complete 32-bit architecture
- Supports 8-bit, 16-bit, and 32-bit data types
- 8 KB unified level-1 cache (code and data)
- Five-stage instruction pipeline:

$\text{IF} \rightarrow \text{ID1} \rightarrow \text{ID2} \rightarrow \text{EX} \rightarrow \text{WB}$

- Complex instructions decoded in two stages
- Single-cycle execution if data exists in cache

9 Memory System

- 32-bit address bus
- Addresses 4 GB of physical memory
- Uses segmentation and paging
- Virtual address space: 64 TB
- Memory organized into four 8-bit wide banks
- Each bank can store up to 1 GB
- Byte enable signals: BE0, BE1, BE2, BE3

9.1 Memory Address Mapping

- Address range: 00000000H to FFFFFFFFH
- Bank selection:
 - 00000000H → Bank 0
 - 00000001H → Bank 1
 - 00000002H → Bank 2
 - 00000003H → Bank 3

10 Input/Output System

- Uses isolated I/O system
- 64 KB I/O address space (0000H–FFFFH)
- 32-bit I/O system divided into four banks
- Most I/O transfers are 8-bit wide
- Disk and video interfaces are 16-bit or 32-bit wide
- Coprocessor I/O range: 800000F8H–800000FFH

11 Processor Versions

- Available versions: 25 MHz, 33 MHz, 50 MHz, 100 MHz
- Advanced versions support clock doubling and tripling

12 Advantages of Intel 80486

- Integrated FPU and cache
- Faster instruction execution

- Improved memory and I/O performance
- Efficient pipeline architecture
- Enhanced support for high-level software

13 Major Differences Between Intel 80386 and Intel 80486

Feature	Intel 80386	Intel 80486
Processor Architecture	32-bit microprocessor	32-bit enhanced microprocessor
Floating Point Unit (FPU)	External coprocessor (80387) required	Built-in floating point unit
Cache Memory	No on-chip cache	8 KB on-chip unified cache
Instruction Pipeline	Limited pipelining	Five-stage instruction pipeline
Instruction Execution	Multiple cycles per instruction	Single-cycle execution for cached instructions
Clock Speed	Up to 33 MHz	Higher clock rates (25–100 MHz versions)
Performance	Lower compared to 80486	Significantly higher due to cache and pipelining
Integration Level	CPU only	CPU, cache, and FPU integrated on one chip